

KAVEESHA SEWMINI

SOFTWARE ENGINEER

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👤 Kaveesha Sewmini



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Gampaha, Sri Lanka



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ABOUT ME

Software Engineering undergraduate with hands-on experience in full-stack development, machine learning, and DevOps based systems. Skilled in building scalable web applications using React, Node.js, Flask, Laravel, and Docker, with practical experience in AI driven and data intensive projects. Passionate about designing efficient, secure, and user centric software solutions, with strong interests in backend systems, cloud deployment, and applied machine learning.

EDUCATION

Sabaragamuwa University of Sri Lanka | 2023 - Present (Expected Graduation: 2027)

• **BSc (Hons) in Software Engineering**

Current GPA: 3.729/4.000

ESOFT Metro Campus | 2022 - 2023

• **Diploma in Information Technology (Pearson Assured)**

Distinction Pass

Programming with Python | Database with SQL | Programming with C#

• **Diploma in English (Pearson Assured)**

Distinction Pass

Rathnavali Balika Vidyalaya, Gampaha | 2013 - 2021

• **G.C.E. Advanced Level Examination - Physical Science (2019 - 2021)**

Z-Score - 1.4281

Physics - A | Mathematics - B | Chemistry - B

• **G.C.E. Ordinary Level Examination - 2018**

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EXPERIENCE

Kubegraf.io – Software Engineer (Part-Time)

December 2025 – March 2026

- Performed **functional and regression testing** for the Kubegraf.io website and SaaS platform, **identifying, documenting, and validating software defects** to ensure product quality and reliability.
- Collaborated with developers to identify, fix, verify, and troubleshoot software defects** and improve application stability through continuous testing and quality assurance practices.
- Contributed to feature development** by proposing usability-focused improvements and validating new functionalities to optimize the user experience.
- Used **Git-based version control** for collaborative development, change tracking, and project contribution management.

PROJECTS

TaskFlow – Full-Stack Task Management Platform (Individual Project)



- Developed and deployed a full-stack task management platform using **React, Vite, Node.js, Express**, and **MongoDB (Mongoose)**.
- Built productivity features including **task CRUD, dashboard analytics, mood tracking, focus streaks**, and **AI-powered task planning using Google Gemini API**.
- Implemented secure authentication and account management using **JWT, bcryptjs, cookie-parser, email verification**, and **crypto-based OTP password reset**.
- Designed and deployed a scalable cloud architecture by hosting the backend on **AWS Lambda** and the frontend on **Vercel**.
- Automated deployment workflows using GitHub **Actions CI/CD** and optimized application delivery with **Docker-based containerization**.

Maternal & Child Health Management System (Group Project) – Team Leader

(Ongoing)

- Leading a team in developing a full-stack healthcare management platform using **React, TypeScript, Node.js, Express, PostgreSQL**, and **Prisma** to digitize manual records and streamline maternal and child health record management for the MOH Office, Imbulpe, Sri Lanka.
- Improved usability and operational efficiency through responsive UI development using **React, Vite, Tailwind CSS**, and **Radix UI**, enabling effective patient data management for healthcare staff.
- Enhanced system security and data integrity using **JWT authentication, encrypted data handling, RESTful APIs, Zod validation**, and **Prisma ORM with PostgreSQL**.

Brain Tumor Classification using MRI Images (Individual Project)



- Improved diagnostic automation by developing an end-to-end medical image classification system using **Python** and **TensorFlow/Keras (custom CNN and VGG16 transfer learning)** to classify MRI scans.
- Enhanced model reliability by training and evaluating models using the **Kaggle MRI dataset**, **NumPy preprocessing**, and **Scikit-learn metrics**.
- Built a full-stack inference system using **Flask**, **TensorFlow**, **React**, and **Axios**, enabling MRI image upload, **preprocessing**, JSON response, and result visualization.
- Ensured seamless deployment and integration by using **Flask CORS**, **Pillow**, **h5py**, and **.h5 model serialization**, enabling stable frontend-backend communication and model serving.

TruthLab - AI-based Fake News Detector (Group Project)



- Built a full-stack fake news detection system using **React (Vite, Tailwind CSS)**, a **Flask REST API**, and **MongoDB Atlas (PyMongo)** with authentication and verification history tracking.
- Enhanced prediction performance by integrating an ML pipeline using **scikit-learn** and a **trained .pkl model (joblib)** to classify news as REAL or FAKE from text or URLs.
- Strengthened backend security and processing by implementing backend services using **Flask Blueprints**, **JWT**, **bcrypt**, **Flask-CORS**, **Requests**, and **BeautifulSoup4** for secure processing and article extraction.

University Sports Management System (Group Project) - Team Leader



- Developed a full-stack web app using **Laravel 12**, **PHP**, **Blade**, **Eloquent ORM**, and **MySQL** for a university sports education portal.
- Improved UI responsiveness and interactivity using **Bootstrap**, **jQuery**, **Axios**, **Select2**, and **Vite** to create a responsive UI with interactive features.
- Strengthened backend architecture using the **Laravel MVC pattern**, **routing**, **middleware**, **sessions**, **queues**, and storage system for scalable architecture.

SKILLS

Technical Skills

- Languages:** Java, JavaScript, TypeScript, Python
- Frontend:** React, Angular, HTML5, CSS3, Bootstrap, Tailwind CSS
- Backend:** Node.js, Express.js, Flask, PHP (Laravel), Spring Boot
- Databases:** MongoDB (Mongoose), MySQL, PostgreSQL
- Machine Learning & AI:** Python, TensorFlow/Keras, Scikit-Learn, NumPy, Pandas
- DevOps & Tools:** Docker, Docker Compose, Kubernetes, Git, GitHub, CI/CD, AWS EC2, AWS Lambda, AWS ECS, AWS S3, Postman, Figma, Adobe Photoshop, JUnit
- APIs & Auth:** RESTful APIs, JWT Authentication, Axios

Soft Skills

- Problem-solving, Team collaboration, Leadership, Communication, Time Management, Adaptability

EXTRACURRICULAR ACTIVITIES

- Student Branch Ambassador - IEEEExtreme 19.0
- Project Chair - SabraXtreme 3.0
- 2nd Year Student Ambassador - Innovation Nation Sri Lanka 2025
- Volunteer - STEMUP Educational Foundation (AI for Beginners - Design Team)
- Volunteer Graphic Designer - IEEE Day Celebration 2025
- Volunteer Graphic Designer - GeekZone v1.0
- Volunteer Graphic Designer - IEEE Day Celebration 2024

CERTIFICATIONS

- Postman API Fundamentals Student Expert
- JavaScript (Intermediate) - HackerRank
- Python for Beginners - CODL - University of Moratuwa
- Bootstrap Mastery: Build Responsive Websites Like a Pro
- Figma for User Interface and User Experience UI/UX Design
- Kubernetes From 10000 Feet
- Python Programming Mastery: From Beginner to Pro
- DevOps Foundations - LinkedIn Learning

REFERENCES

- Prof. B.T.G.S. Kumara** (Dean, Faculty of Computing, Sabaragamuwa University of Sri Lanka)
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- Ms. H.M.C. Nirmani** (Lecturer (Probationary), Department of Software Engineering, Faculty of Computing, Sabaragamuwa University of Sri Lanka)
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